

[illegible]

Lysine methylation creates binding sites for co-repressors

MT1A (Suv39h) is one of 7 human enzymes that can methylate H3K9

Bound by chromatin proteins

H3 ARTEQTARKSTGGKAPRRQLATKAARRSA

HP1

Heterochromatin silencing

PC1 (PRC1 complex)

Polycomb silencing

MT1B (E2f2) is in the PRC2 complex

SUZ12

ESC

An activation paradigm

ON

HAT

Ac

read acetyl marks (X) to turn genes on

erase acetyl marks (HDACs) to silence genes

A silencing paradigm

OFF

HMT

Me

read methyl marks with effectors (Y) to silence genes

erase pre-existing acetyl marks (HDACs)

write new methyl marks (HMTs)

read methyl marks with effectors (Y) to silence genes

Inactive X chromosome lacks H4 acetylation

anti-acetylH4

immunofluorescence (IF) staining of metaphase chromosomes from female mammalian cells

Two gene silencing systems: selective use of lysine (K) methylation

H3 N-ARTKQTARKST

• OFF methyl groups

KAARKSAPA..

α-H3K9me

α-H3K27me

HP1 = X

Pc = Y

always off

regulated off

Des:gn

对人类样本进行单核转录组 (snRNA-seq) 和表观基因组 (snATACseq) 分析, 旨在研究C9-ALS或C9-FTD携带者两个关键皮层区域 (运动皮层和额叶皮层) 的细胞类型特异性分子改变, 以揭示区域差异。

GO 分析

从生物过程 (如细胞分裂)、分子功能 (如酶催化)、细胞组分 (如细胞核定位) 三个层面, 描述基因功能, 梳理基因参与的具体生命活动。

KEGG 分析

基于转录组数据库 (代谢、信号转导、疾病相关通路等), 挖掘基因参与的通路, 解析 "基因在生物系统中如何协同作用"。关联基因与生理/病理通路 (如肿瘤信号通路), 揭示基因在疾病、代谢中的调控关系。

治疗研究策略

核心靶点: 运动神经元 (Motor Neurons)

干预策略:

药理学 (Pharmacological):

神经营养 / 保护 (NGF and neuroprotection), 抗凋亡 (Anti - apoptotic) 抗炎 (Anti - inflammatory), 抗兴奋毒性 (Anti - excitotoxicity), 抗氧化 (Anti - oxidant), 抗蛋白聚集 (Anti - aggregation)

非药理学 (Non - pharmacological): 细胞治疗 (Cell - based therapies)

ChIP Tech for Transcription

1. Single Gene Expression Test

RT-qPCR

RNA sample

1. Amplification efficiency for relative quantification

1. Total RNA

2. DNase to remove DNA

3. Reverse transcription

4. PCR

5. Gel electrophoresis

6. DNA sequencing

7. Data analysis

8. Report

可以验证哪些基因在特定细胞中表达

PAGE or kit 富集 small RNA + Adapter. PolyA 捕获 mRNA → Fragment

Reverse transcription-PCR

RT-PCR

1. Isolate RNA

2. Anneal Oligo(dT) primers

3. First-strand synthesis

4. First cycle PCR synthesis

5. PCR amplification

Reverse transcription quantitative real-time PCR (qPCR)

1. Isolate DNA

2. Denaturation

3. Primer annealing and Extension

4. Denaturation

5. Primer annealing and Extension

6. DNA synthesis and fluorescence detection

Unbound dye

Bound dye

cross-link chromatin to stabilize protein:DNA interactions

ChIP

shear to small pieces

precipitate with antibody

Purify DNA and detect by hybridization or sequencing

ChIP Transcriptome

转录组 (transcriptome)

定义: 细胞、组织、生物个体中, 所有 RNA 分子的集合 (mRNA, miRNA, lncRNA)。

本质: 反映特定状态下基因的活跃程度

RNA 全生命周期解析

覆盖 RNA 的转录、表达水平、功能、定位、转运、降解全过程;

关注转录本结构: 启动子起始位点、5'/3'末端序列、可变剪接模式、转录后修饰 (如甲基化、加帽加尾)

现代转录组学技术与应用

依赖高通量方法 (如 RNA - Seq), 实现多转录本、生理/病理状态的并行分析

Combinations of marks encode epigenomic state

Enhancers

H3K4me1

H3K27ac

DNase

Promoters

H3K4me3

H3K9ac

DNase

Transcribed

H3K36me3

H3K79me2

H4K20me1

Repressed

H3K9me3

H4K27me3

DNAmethyl

chromatin fiber

DNA

nucleus

nucleosome

H3K4me3

H3K4me1

H3K27ac

H3K36me3

H4K20me1

H3K79me2

H3K27me3

H3K9me3

H3K9ac

H3K18ac

揭示转录组与表型的关联: 从细胞到个体层面, 解析基因表达模式如何决定生物性状、疾病发生。

精准分析细胞特性: 基于深度 RNA - Seq, 识别特定细胞类型中差异基因 / 转录本的表达模式 (如肿瘤细胞与正常细胞的 mRNA 差异、神经细胞特有的 lncRNA)

转录组: 所有的 RNA

转录组: 和 RNA 有关的一切

包括所有 RNA

基因表达调控流程:

表观遗传 (Epigenetics): DNA 甲基化、组蛋白修饰影响基因开启 / 关闭;

转录 (Transcription): 转录因子结合增强子 / 启动子, 调控 RNA 聚合酶, 启动转录;

转录后 (Post - transcription): RNA 加帽、加尾、可变剪接, 加工成成熟转录本;

翻译沉默 (Silencing): miRNA 结合靶 mRNA, 抑制翻译过程;

翻译 (Translation): 信号肽 (蛋白 N 端) 引导蛋白运输, 后被剪切;

翻译后修饰 (Post - translational modification): 蛋白共价修饰 (磷酸化、乙酰化等) 改变功能

2. Microarray

How do all the different genes act together to produce and maintain this functioning organism?

this is a grid containing 7many or even all of the genes of an organism

the cDNAs are labeled with fluorescent tags

single-stranded DNA probes tethered to a solid support

these represent a particular gene

compare gene expression in a regular cell and a cancer cell

tissue 1 → red

tissue 2 → green

both → yellow

Lower reagent cost

Can be done in your own lab

Advantages

Requires prior knowledge of gene/mRNA

Limited number of probes/targets Disad..

3. RNA - Seq

Workflow of RNA-seq by NGS

RNA isolation

rRNA depletion

Library construction

Deep sequencing

Bioinformatics

hybridization with biotin-labeled anti-rRNA probes

selective degradation by a 5'-3' exonuclease that specifically recognizes rRNA

Single-end (SE) sequencing is cheaper than PE

Paired end (PE) sequencing improves mapping to repeat sequences in the genome

PE sequencing also improves mapping across exon-exon junctions

PE improves accuracy for detection of differential expression for low-expressed genes

Reverse transcription quantitative real-time PCR (RT-qPCR)

1. Isolate RNA

2. Anneal Oligo(dT) primers

3. First strand synthesis

4. Denaturation

5. Primer annealing and Extension

6. DNA synthesis and fluorescence detection

Unbound dye

Bound dye

Annotated transcripts

Gene 1, Isoform A

Gene 1, Isoform B

Types of transcript novelty

Prefix

Suffix

Known

Incomplete splice match (ISM)

Novel in catalog (NIC)

Novel not in catalog (NNC)

Genomic

Antisense

异构体新颖性分类 (Isoform novelty categories)

已知转录本 (Known): 与数据库注释完全匹配的转录本。

不完全剪接匹配 (ISM): 分 Prefix (5'端延伸不全)、Suffix (3'端延伸不全), 是已知基因的 "不完整版本"。

目录中新颖 (NIC): 匹配已知基因, 但剪接模式 / 结构在数据库中无记录。

目录外新颖 (NNC): 来自已知基因区域, 但整体结构、剪接均未记录。

基因组新颖 (Genomic): 非编码区转录, 无已知基因注释。

反义链 (Antisense): 与已知基因反义链转录, 可能调控基因表达。

TALON 流程: 长读长转录组分析 "标准化 pipeline"

核心作用:

处理 PacBio, 纳米孔等长读长测序数据, 识别新转录本、校正错误剪接、精准解析异构体 (isoform)

关键步骤: 利用 PacBio 和 Nanopore 不同 cell line 的转录组数据, 用 TranscriptClean 校正测序误差 (如非经典剪接位点); 转录本注释: 结合已知数据库 (如 GENCODE), 标记已知 / 新转录本、量化表达;

验证: 用生物学重复验证不可靠新转录本 (需稳定出现, 无内部引物干扰);

输出结果: 生成定制化转录组 GTF 文件 + 表达定量矩阵 (talon_abundance)

PacBio 优势: 对已知转录本、新转录本的 5'/3'完整性支持率更高

纳米孔特点: 对反义链 (Antisense)、基因间区 (Intergenic) 转录本支持率相对高, 但整体末端完整性弱于 PacBio

3. Amyotrophic lateral sclerosis ALS

肌萎缩侧索硬化症 (ALS) 是一种影响大脑和脊髓神经细胞的神经系统疾病, 该病会导致肌肉控制能力丧失, 并随时间推移持续恶化。

该疾病的确切病因至今未明, 仅少数病例与遗传相关。

典型症状初期表现为单侧肌肉无力、吞咽困难或言语含糊。

病情进展将最终损害运动、言语、进食及呼吸所需的肌肉控制功能。

目前尚无针对这一致命性疾病的治愈方案

RNA 结合蛋白 TDP-43 在 neurons 中 ALS 中聚集

C9orf72 中重复序列扩增

microRNAs

21-28 nt long

UGAGUAGUAGUUGUAGUUGUUGU

let7a-1

ENCODER4: sequence microRNAs

First Half

Second Half

Shad read capturing falls short of identifying mRNA isoforms

short read

long read

messenger RNAs

300-110,000 nt long

Most annotated mRNAs less than 5kb

ENCODER4: full-length transcript isoform

Variant 1 (NM_001205505)

Variant 2 (NM_001205505)

Variant 3 (NM_001205505)

Variant 4 (NM_001205505)

Variant 5 (NM_001205505)

Variant 6 (NM_001205505)

Variant 7 (NM_001205505)

Variant 8 (NM_001205505)

Variant 9 (NM_001205505)

Variant 10 (NM_001205505)

Variant 11 (NM_001205505)

Variant 12 (NM_001205505)

Variant 13 (NM_001205505)

Variant 14 (NM_001205505)

Variant 15 (NM_001205505)

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Variant 96 (NM_001205505)

Variant 97 (NM_001205505)

Variant 98 (NM_001205505)

Variant 99 (NM_001205505)

Variant 100 (NM_001205505)

Short reads, suitable for gene expression, resequencing

Longer reads, suitable for new transcriptomes, WGS for eukaryotes

Longest* reads

For Amplicon Seq, Metagenomics

Quality assessment (filter low quality reads)

Unmapped reads

De novo assembly

Map reads to a reference

Junction library

Transcriptome map (quality and quantity)

Normalization

Novel transcript and isoform detection

Non-coding RNA identification

Fusion gene detection

SNP discovery

Copy number variations identification

Biological insight

Normalization:

RPKM: (reads mapped to a gene) / (#(Total mapped reads) * 10^6)

RPKM: (reads per kilobase of transcript per million mapped reads):

Gene length in kb

TPM: (reads mapped to the i - th gene) / (sum of reads mapped to all genes) * 10^6

TPM: fragments per kilobase of transcript per million mapped fragments

双端测序: one fragment - two reads

TPM: Transcripts / million transcripts

Ti = (Length of the i - th gene (kb) * sum of reads mapped to all genes) / (sum of reads mapped to the i - th gene)

Microarray vs. RNA-Seq

Limited probeset based on prior knowledge of the transcriptome

Higher throughput

Analysis is more user-friendly than RNA-Seq currently

Comprehensive overview of the transcriptome

Best dynamic range

Gene fusion and isoform discovery 单碱基分辨率

Can detect SNVs

SMART-based

Fresh tissue or frozen single cell suspension

One-cell at a time, lower throughput

Full length

Lower dropout

Transcripts / cell higher

Per cell transcription more accurate

Dropout-based

Fresh tissue or frozen single cell suspension

All droplets together, higher throughput

Cell barcode

UMI for PCR bias correction

3' (or 5') bias

Higher dropout

Transcripts / cell lower

Per cluster transcription more accurate

4. Single-Molecule RNA seq

5. sc RNA-seq

Visium 平台 (10x Genomics) 细胞直径: 6.5um

核心设计: 基于荧光 (array-based) 的空间转录组技术, 样本区域 6.5mm*6.5mm, 含 4992 个六边形排列的捕获点 (spot)

Spot 参数: 直径 55um, 中心间距 100um; 每个 spot 含数百万探针, 带空间条形码 (spatial barcode)、polyT 序列, 特异性捕获 poly (A) + mRNA

分辨率特点: 每个 spot 对应 1 - 10 个细胞, 非单细胞分辨率, 但可实现转录组的空间定位

ST 平台 (Spatial Transcriptomics)

核心设计: 玻璃载玻片 (slide) 上分布捕获探针阵列, 不同版本参数有差异 (如 1 代 1000 个 spot, 直径 100um; 2 代 2000 个 spot)

探针结构: 含 polyT 序列, UMI (Unique Molecular Identifier, 用于定标), 空间条形码 (标记 spot 位置), 扩增及测序接头, 切割位点, 实现 mRNA 捕获与定位关联

Chao-1 Single-cell Transcriptome & Tech

分离、微流控、稀释、流式

SMART-based

Fresh tissue or frozen single cell suspension

One-cell at a time, lower throughput

Full length

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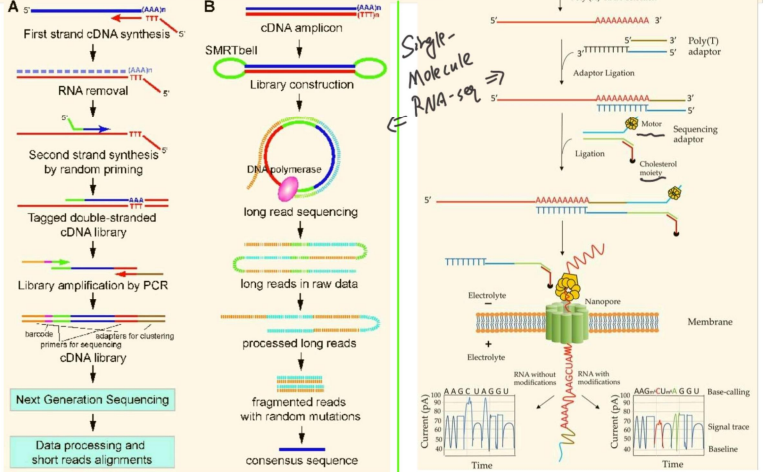
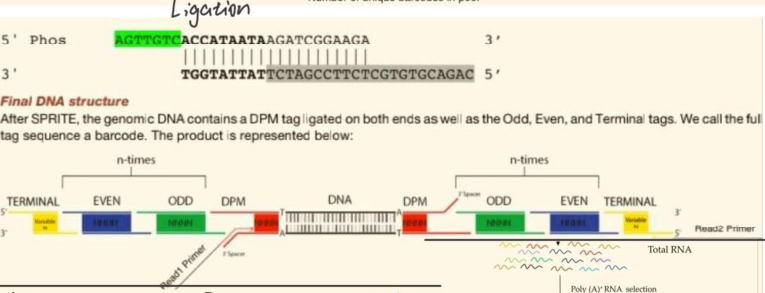
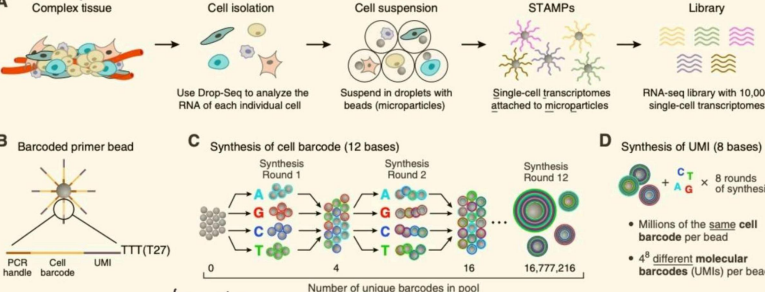
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Chao-1 Single-cell Transcriptome & Tech

分离、微流控、稀释、流式

Barcoding Methods: Synthesis



空间转录组核心技术

1. 组织制备 (Tissue Preparation)

流程: 固定组织→H&E 染色成像 (保留形态) →透化处理 (让 mRNA 释放并结合探针) →mRNA 捕获 (探针通过 polyT 结合 poly (A) + mRNA) →反转录合成 cDNA→移除组织→切割 cDNA

2. 文库构建与测序 (Library Preparation & Sequencing)

流程: 带空间条形码的 cDNA 混合 pooling→用 Slide cassette 分隔反应腔室→构建文库后测序; 测序读取双端信息: Read1 读空间条形码 + UMI, Read2 读基因序列。

3. 质量控制 (QC)

目的: 确保不同组织类型适配实验流程, 尤其是透化时间。

方法: 用无空间条形码的 oligo (dT) 引物预实验, 结合荧光成像验证 mRNA 捕获效率, 优化实验参数 (如冷冻程序、切片厚度、染色方案、RNA 完整性值 RIN)

4. 数据解析

关键差异: 无空间条形码 dsDNA, 不用起始蛋白和 nickase; 不产生 displaced DNA, 无切割步骤。

核心酶: Phi29 DNA 聚合酶 (来自枯草芽孢杆菌), 特点是 高持续合成能力、高保真度、低错误率, 可生成超长 DNA 片段。

产物: 形成连续的多拷贝串联体 (concatemers) —— 同一 DNA 片段首尾相连的长链, 最终组装成 DNA 纳米球 (DNB) (含 300 - 500 个拷贝)

5. Nanoball Seq

高通量测序全基因组序列, 微环扩增技术扩增小片段

6. Nanoball Seq

高通量测序全基因组序列, 微环扩增技术扩增小片段

文库制备 (DNA Nanoball Library Preparation)

基因组 DNA 提取: 标准要求 1μg, 片段 >23kb; 片段化: 片段筛选: 选 100 - 300bp (最优) 的片段;

末端修复: 修复 DNA 断裂末端; 接头连接与 PCR 扩增: 给片段加测序接头, PCR 扩增; → adapter 成环

环化单链 DNA (ssCirDNA): 连接拆分寡核苷酸, 变性后形成环状单链; 改造版滚环扩增: 用 Phi29 酶扩增, 形成 DNA 纳米球 (DNB)。

2. 测序 (Loading Onto Flow Cell/Sequencing)

加载纳米球: DNB 加载到带图案的流动槽 (flow cell), 每个纳米球结合 1 个带正电的氨基硅烷;

测序原理: 类似 Illumina, 通过 组合探针锚定合成 (cPAS) 技术, 用 DNA 聚合酶掺入荧光标记 dNTP (含终止子), 洗涤未结合的 dNTP, 成像后切割染料和终止子, 循环完成测序

核心优势:

低扩增偏差: 线性扩增 (RCA 是线性, PCR 是指数), 无误差累积;

高信号强度: 单个 ssCirDNA 信号弱, DNB 含多拷贝, 信号强, 提升碱基识别准确性;

精准定量: UMI (Unique Molecular Identifier) 结合 RCA, 降低 PCR 偏差; 混合测序深度 高精准度测序 (如人全基因组 单细胞测序)

